

QRC Tech File

Monday, September 21, 2026 11:26 AM

1

What is a QRC Technology File (qrcTechFile) ?

A qrcTechFile is a foundry-provided technology file (binary) that contains process technology data used by Quantus/QRC for parasitic extraction (R, C, and related parameters).

1 Definition & Purpose

- QRC Technology File (qrcTechFile) is a **binary technology file** provided by the **foundry**.
- It contains **process technology data** required by Quantus/QRC to calculate **parasitic resistance (R)**, capacitance (C) and other parasitic effects.
- It models the electrical behavior of the actual silicon process (metal layers, vias, dielectrics, etc.).
- Without this file, the extractor (QRC/Quantus) cannot compute accurate parasitics.

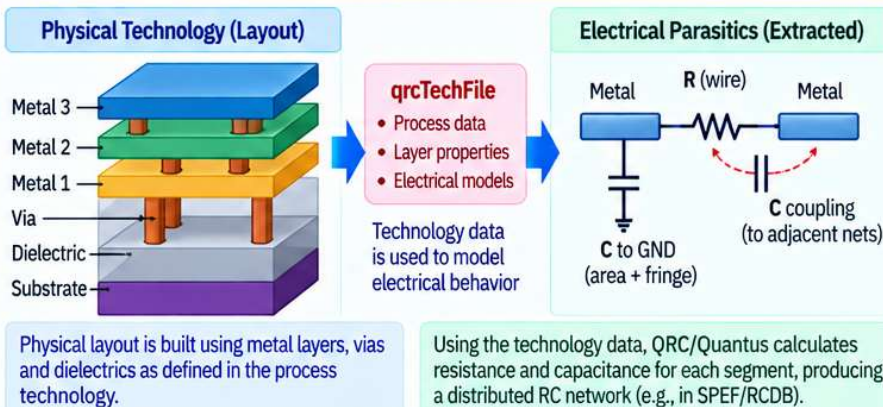
2 Why Quantus/QRC Needs It ?

- QRC/Quantus uses the technology data to:
 - Calculate resistance of wires and vias
 - Calculate ground and coupling capacitances
 - Apply correct process-dependent models
 - Model 2D/3D parasitic effects
 - Support multiple RC corners (best/typ/worst)
- It ensures the extracted parasitics match the electrical behavior of the actual foundry process.

3 Foundry-Provided Technology File

- qrcTechFile is generated automatically by the foundry using their process data and models.
- It is not created manually by the user.
- It goes through foundry validation (e.g., 28nm, 14nm, 7nm).
- Different RC corners (best/typ/worst) are provided.
- It ensures accurate and consistent parasitics across the same PDK.

4 Relationship: Physical Technology → Electrical Parasitics



5 What Information Does It Contain ?

- Metal layer information (thickness, width, spacing, permittivity, etc.)
- Sheet resistivity for each layer
- Via resistance and via models
- Dielectric properties (perm, constant, thickness)
- 2D/3D capacitance models (ground, coupling, fringe, area)
- Layer-to-layer coupling parameters
- Process-dependent parameters (PVT, RC corners)
- Special structures (e.g., fill, density effects, etc.)



Key Takeaways:

- qrcTechFile is a foundry-provided binary file containing process technology data for parasitic extraction.
- It enables Quantus/QRC to calculate accurate R and C values based on the real silicon process.
- It is different from LEF/DEF – LEF/DEF provide physical geometry, while qrcTechFile provides electrical models.
- Always use the correct qrcTechFile for the technology node and RC corner specified by the foundry.

2

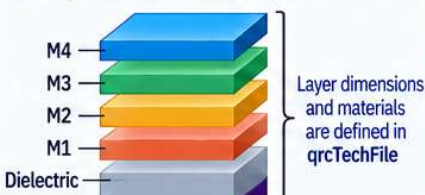
QRC Tech File (Part 2)

What information is actually stored in a QRC Technology File ?

A qrcTechFile contains process technology data that defines how interconnects behave electrically, including resistance and capacitance models for accurate parasitic extraction.

1 Metal-Layer Information

- Number of metal layers (M1, M2, ...)
- Layer thickness, width, spacing
- Layer material (e.g., Cu, Al)
- Dielectric above/below each layer
- Layer permittivity and height from substrate
- Layer type (routing, special, etc.)



2 Sheet Resistivity

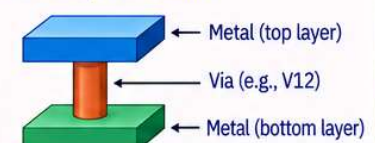
- Sheet resistance ($\Omega/\square$) for each metal layer
- Process and temperature dependent values
- Different values for different corners (best/typ/worst)

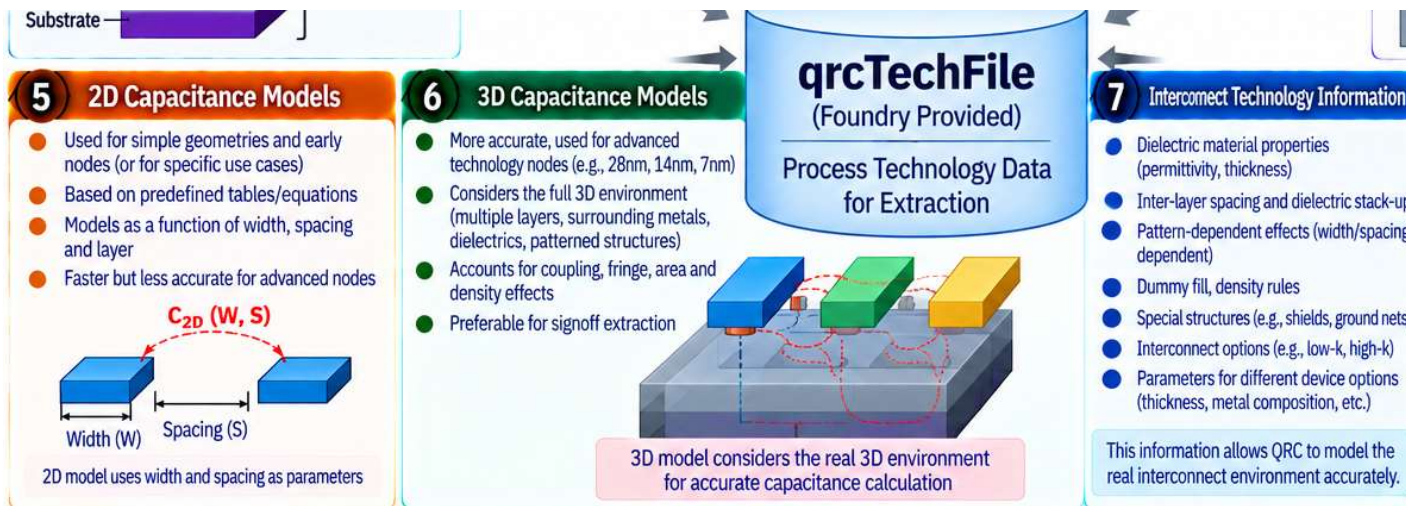
$$R = R_{sheet} \times \frac{L}{W}$$

R_{sheet} value for each metal is stored in qrcTechFile

3 Via Resistance

- Via resistance for each via type (e.g., V12, V23, V34, ...)
- Resistance depends on via size, cut number and process
- Includes stacked vias
- Corner-dependent values





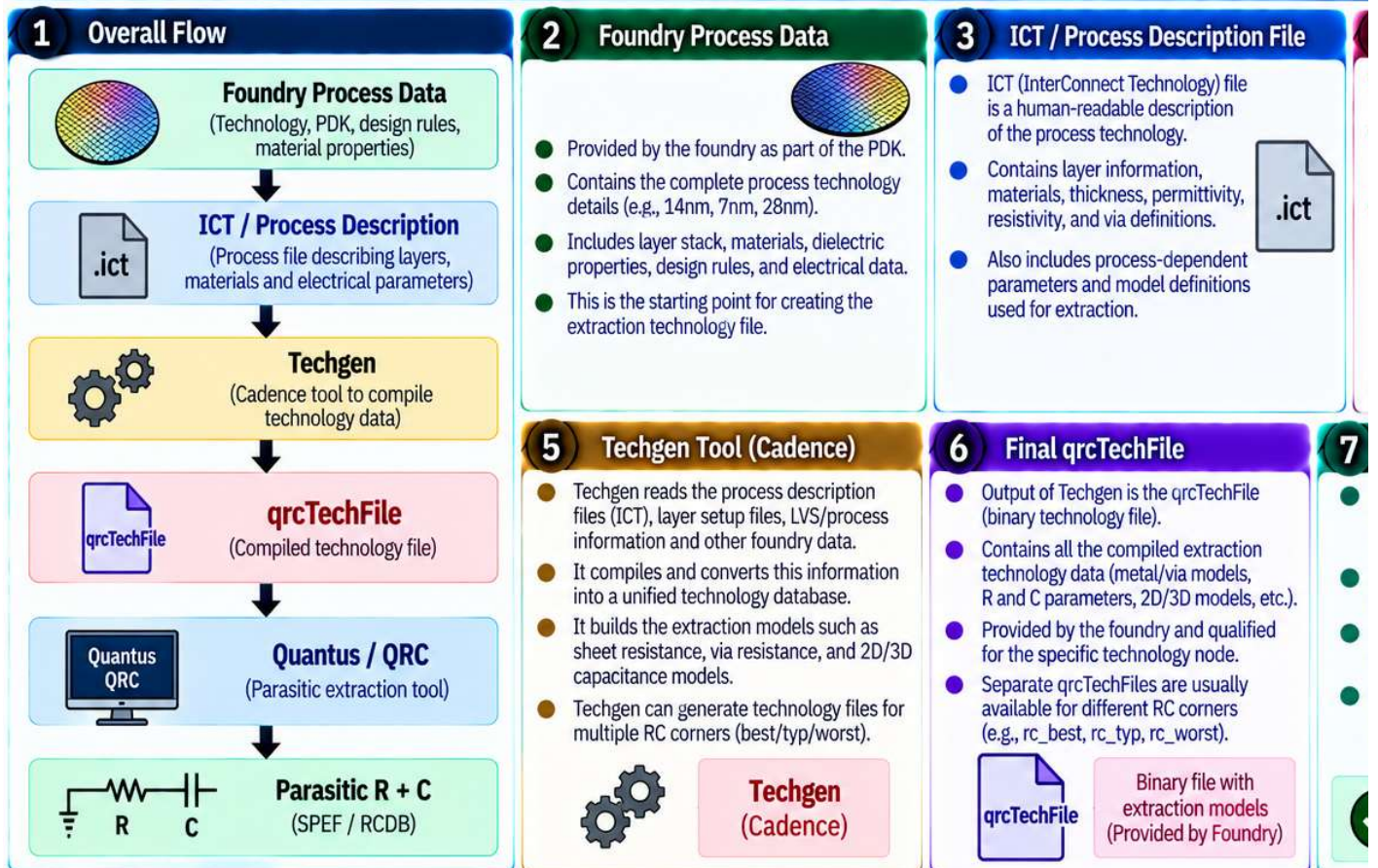
Key Takeaways:

- qrcTechFile contains all technology data needed to calculate parasitics (R, C) from the physical design.
- It includes metal/via information, resistivity, 2D/3D capacitance models and process-dependent parameters.
- It is provided by the foundry and is essential for accurate extraction and signoff.

3

How is a QRC Technology File Created ?

The qrcTechFile is generated by Cadence Techgen using foundry process data (e.g., ICT, layer setup). It is a compiled technology file that contains the extraction models used by Quantus/QRC.



Key Takeaways:

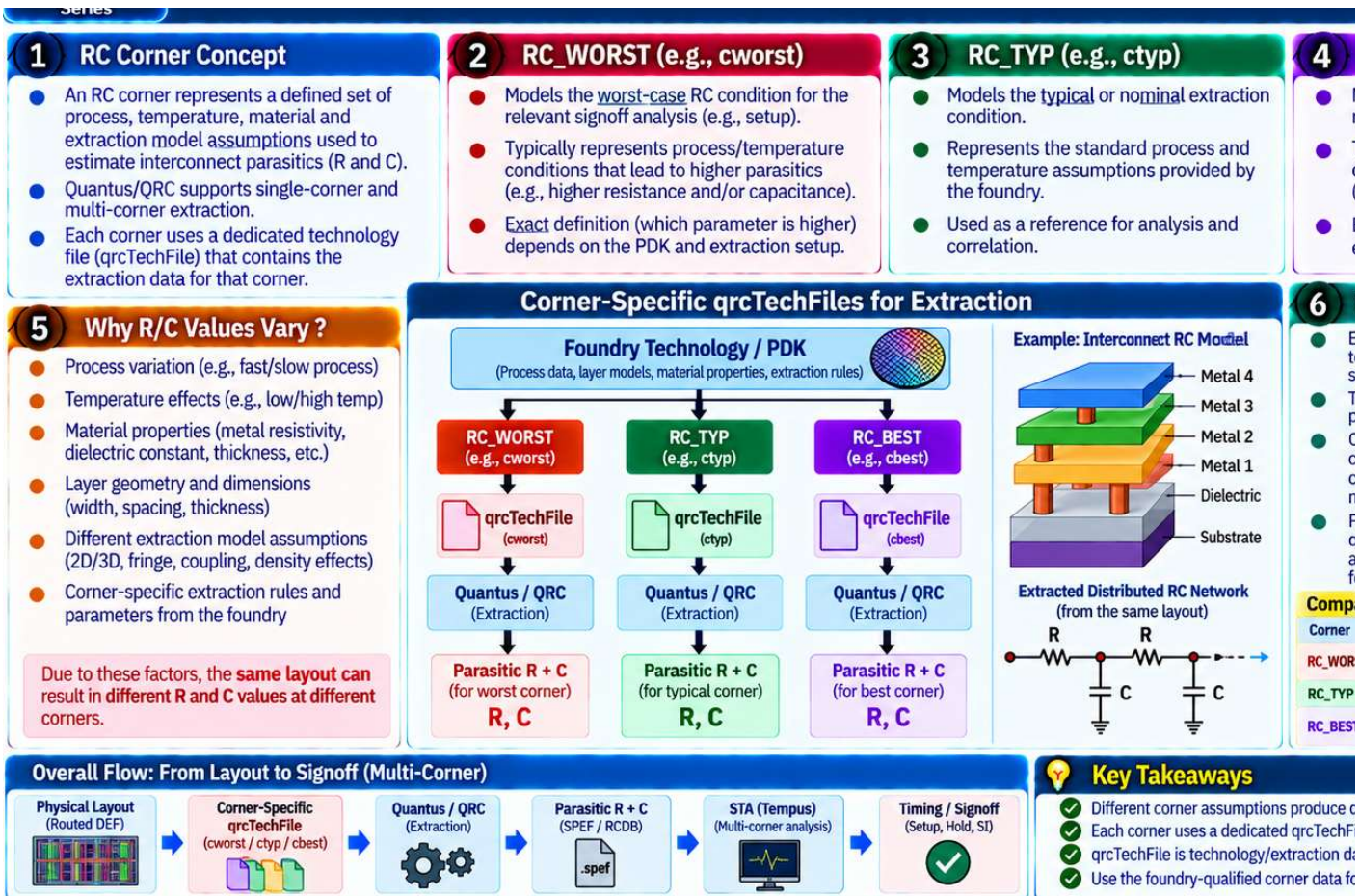
- qrcTechFile is created using foundry process data through Cadence Techgen.
- Inputs include ICT file, layer setup, LVS/process information and other technology files.
- It is a compiled binary file containing extraction models (R, C, via, etc.).
- PD engineers usually use the foundry-provided qrcTechFile, not create it manually.

4

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Why do we need a different QRC Technology File for different RC co

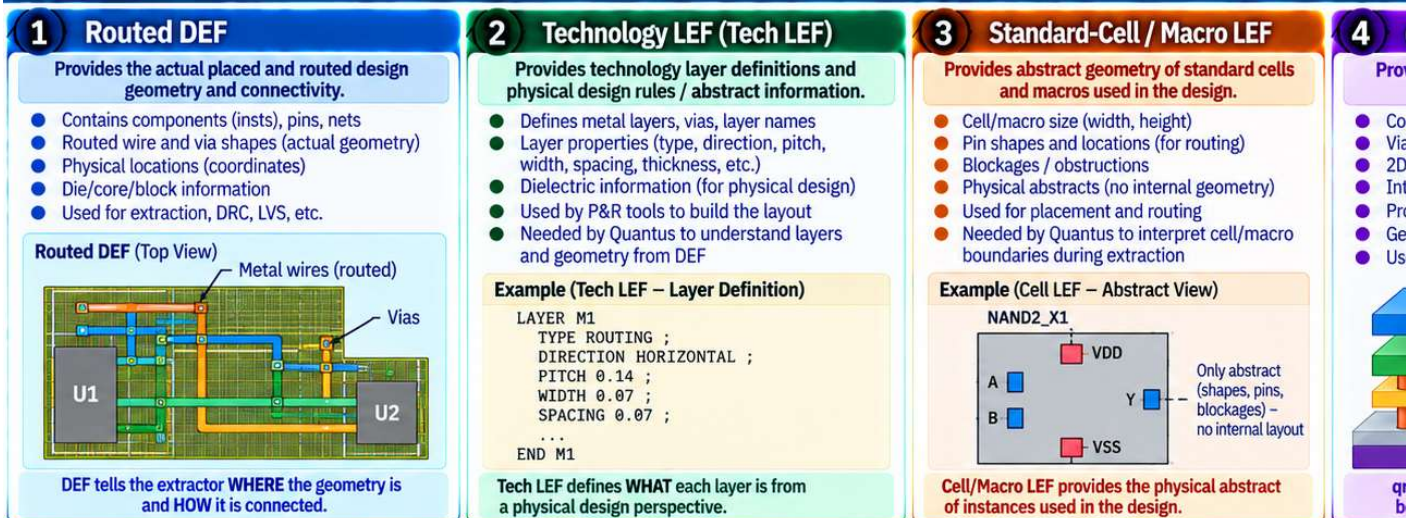
The same physical layout is extracted under different process/corner assumptions, producing different parasitic R



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What is the relationship between QRC Technology File, LEF and

LEF/DEF describe the physical design representation; qrcTechFile provides the extraction technology models used by



Comparison of Inputs

Aspect	Routed DEF	Technology LEF	Standard-Cell / Macro LEF	qrcTechFile
Main purpose	Design geometry & connectivity	Technology layers & rules	Cell/macro abstracts	Extraction technology data (R, C)
Provided by	P&R tool (design database)	Foundry (PDK)	Standard cell / IP vendor (or foundry)	Foundry (PDK)
Information	Insts, nets, routed wires/vias, coordinates	Layer names, type, pitch, width, spacing, thickness, etc.	Cell size, pins, blockages, abstract shapes	Sheet resistivity, via resistance, 2D/3D capacitance models, process data
Used by	Quantus/QRC, DRC, LVS, etc.	P&R tools, Quantus/QRC	P&R tools, Quantus/QRC	Quantus/QRC (extraction)
Nature of data	Design state (layout)	Physical technology description	Physical abstract (no internal details)	Electrical models for extraction

Layer Name Mapping / Corre

The layer names in Tech LEF and qrcTec to the same technology. In some flows, to map extraction layer names to LEF la

Purpose	Tech LEF Layer Name	qrcTechFile Layer
Metal 1	M1	M1
Metal 2	M2	M2
Via 1	VIA1	VIA1

Complete Flow



Important Point:
LEF/DEF tell the extractor **WHAT** and **WHERE** the geometry is; qrcTechFile tells it **HOW** that geometry behaves electrically.

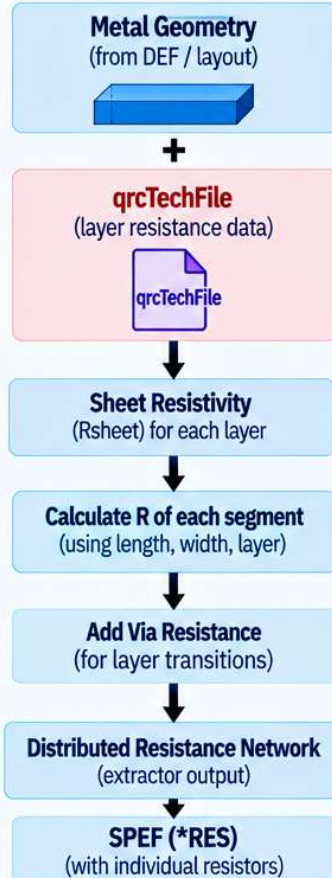
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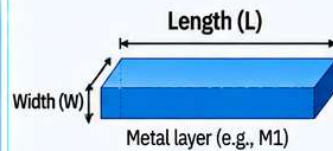
How does Quantus use the QRC Technology File to calculate resistance

Quantus uses the metal geometry from the layout and the layer-wise resistance data from the qrcTechFile to calculate the resistance of each wire segment and via, and outputs a distributed resistance network in

1 Overall Flow



2 Sheet Resistance and Resistance Calculation



Where, R_{sheet} = sheet resistance of the metal layer (from qrcTechFile)
 L = length of the wire segment
 W = width of the wire segment
 N = number of squares

$$R = R_{sheet} \times \frac{L}{W}$$

$$\text{Number of squares (N)} = \frac{L}{W}$$

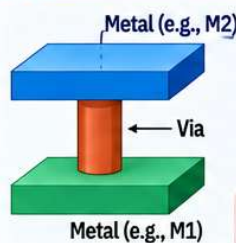
$$R = R_{sheet} \times N$$

3 Layer-Specific Resistance

Metal Layer	Rsheet (Ω) [Example]
M1	0.080
M2	0.070
M3	0.060
M4	0.050
M5	0.040

Exact values are technology dependent (provided in qrcTechFile)

4 Via Resistance



- Each via (e.g., V12, V23, V34) has a specific resistance value.
- Via resistance depends on via size, cut number and process.
- Values are provided in qrcTechFile (for each via type).
- Added in series with metal resistance when moving between layers.

R_{via} (e.g., V12) = value from qrcTechFile (e.g., 0.5 Ω, 1.0 Ω, etc. – technology dependent)

5 Distributed Resistance Network

- Quantus divides each net into small segments based on the layout geometry.
- Calculates resistance for each segment using layer-specific Rsheet.
- Adds resistances at layer transitions (via resistance).
- Generates a distributed RC network (SPEF).
- The resistance part is written in SPEF *RES records.

6 Example: SPEF *RES Section (Resistance)

```

*RES
* II  U1/A      U2/Z
R1    U1/A      N1      0.12 ; M1 wire segment
R2    N1        N2      0.08 ; M1 wire segment
R3    N2        N3      0.45 ; Via V12 resistance
R4    N3        N4      0.10 ; M2 wire segment
R5    N4        U2/Z     0.07 ; M2 wire segment
*END
  
```

- Each segment has a specific resistance value.
- Values are provided in qrcTechFile (for each via type).
- The resistance part is written in SPEF *RES records.



Key Takeaways:

- Quantus uses metal geometry (length, width, layer) and Rsheet from qrcTechFile to calculate wire resistance.
- Via resistances are also taken from qrcTechFile and added at layer transitions.
- The result is a distributed resistance network written in SPEF using *RES.
- Resistance values are layer-specific and technology dependent, so the correct qrcTechFile must be used.
- Accurate resistance extraction is critical for correct timing, IR drop and signal integrity analysis.

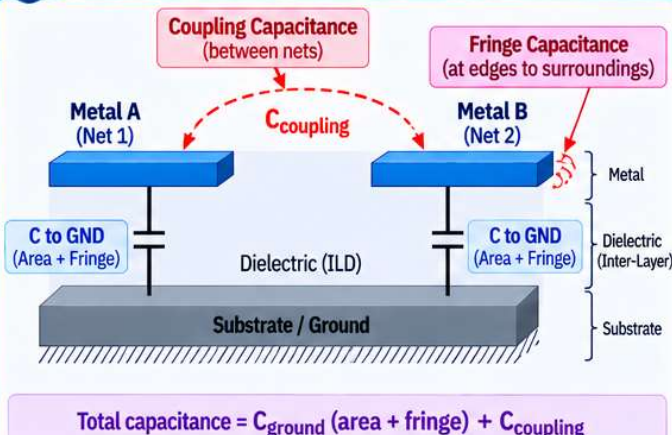
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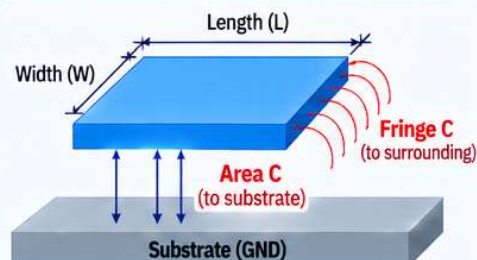
How does Quantus use the QRC Technology File to calculate capacitance

Quantus uses the geometry from the layout (DEF) and the dielectric and capacitance models from the qrcTechFile to calculate ground, fringe and coupling capacitances, considering 2D/3D effects, layer spacing and the surrounding

1 Types of Capacitance in Extraction

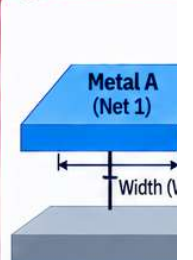


2 Area and Fringe Capacitance



- Area capacitance (C_{area}) depends on the overlap area between metal and the underlying dielectric.
- Fringe capacitance (C_{fringe}) is due to fringing electric fields at the edges of the metal.
- Values are derived from 2D/3D capacitance models in the qrcTechFile, which include dielectric thickness, permittivity and surrounding structures.

3 Coupling



- Coupling capacitance depends on the length, and the spacing between the nets.
- Closer spacing increases coupling capacitance.
- qrcTechFile provides coupling capacitance values based on technology.
- Coupling with ground is also considered.

4 Key Factors Used from qrcTechFile

Parameter	Role in Capacitance Calculation	Source (qrcTechFile)
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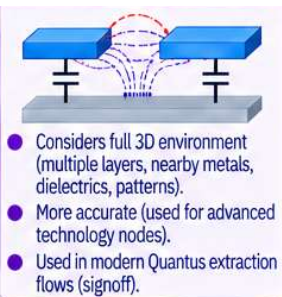
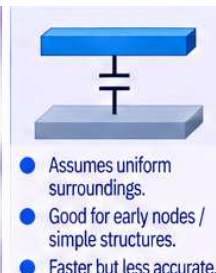
5 2D vs 3D Capacitance Modeling

2D Model (Simple)	3D Model (Advanced)
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6 Relativ

Example: SPEF

Dielectric constant (ϵ_r)	Determines capacitance values between layers and to substrate	Layer dielectric models
Dielectric thickness	Distance between layers / to substrate	ILD stack data
2D/3D capacitance models	Accurate C calculation (ground, coupling, fringe)	Pre-characterized models
Layer-to-layer interaction	Adjacent-layer coupling effects	Coupling tables
Metal width, spacing, thickness	Geometry parameters for C	Layer definitions
Process / corner variation	Different C values for best/typ/worst	Corner-specific data



***CAP**

C1	N1
C2	N2
C3	N1
C4	N4
...	...

***END**

- Quantus writes using *CAP records
- Includes both area and coupling capacitances
- These values are used for timing and signoff



Key Takeaways:

- Quantus uses metal geometry from DEF and dielectric/capacitance models from qrcTechFile.
- It calculates area, fringe, ground and coupling capacitances considering real process data.
- 2D/3D models in the qrcTechFile enable accurate extraction, especially for advanced nodes.
- The extracted capacitances are written in SPEF using *CAP records and used for STA signoff.

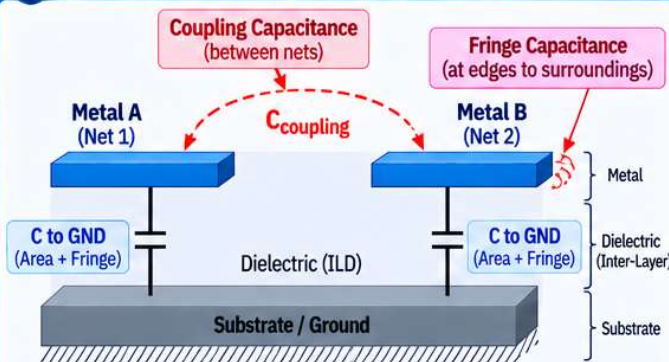
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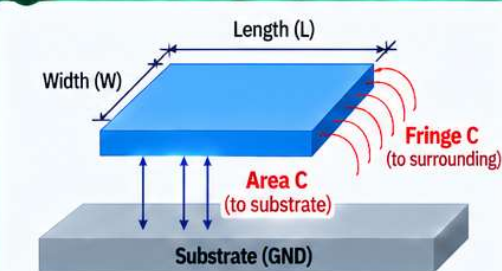
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1 Types of Capacitance in Extraction

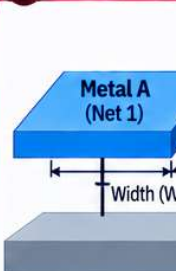


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- Area capacitance (C_{area}) depends on the overlap area between metal and the underlying dielectric.
- Fringe capacitance (C_{fringe}) is due to fringing electric fields at the edges of the metal.
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3 Coupling

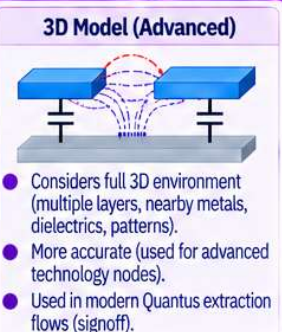


- Coupling capacitance depends on the coupling length, and the spacing between the nets.
- Closer spacing increases coupling capacitance.
- qrcTechFile contains coupling models based on 2D/3D effects.
- Coupling with nearby structures is also considered.

4 Key Factors Used from qrcTechFile

Parameter	Role in Capacitance Calculation	Source (qrcTechFile)
Dielectric constant (ϵ_r)	Determines capacitance values between layers and to substrate	Layer dielectric models
Dielectric thickness	Distance between layers / to substrate	ILD stack data
2D/3D capacitance models	Accurate C calculation (ground, coupling, fringe)	Pre-characterized models
Layer-to-layer interaction	Adjacent-layer coupling effects	Coupling tables
Metal width, spacing, thickness	Geometry parameters for C	Layer definitions
Process / corner variation	Different C values for best/typ/worst	Corner-specific data

5 2D vs 3D Capacitance Modeling



6 Relationship between .ict, .tch and qrcTechFile

Example: SPEF

***CAP**

C1	N1
C2	N2
C3	N1
C4	N4
...	...

***END**

- Quantus writes using *CAP records
- Includes both area and coupling capacitances
- These values are used for timing and signoff



Key Takeaways:

- Quantus uses metal geometry from DEF and dielectric/capacitance models from qrcTechFile.
- It calculates area, fringe, ground and coupling capacitances considering real process data.
- 2D/3D models in the qrcTechFile enable accurate extraction, especially for advanced nodes.
- The extracted capacitances are written in SPEF using *CAP records and used for STA signoff.

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Topic 8

What is the difference between .ict, .tch and qrcTechFile?

These files are related to technology information used for parasitic extraction. The .ict file is used by Techgen to create a technology file, which is provided by the foundry as qrcTechFile.

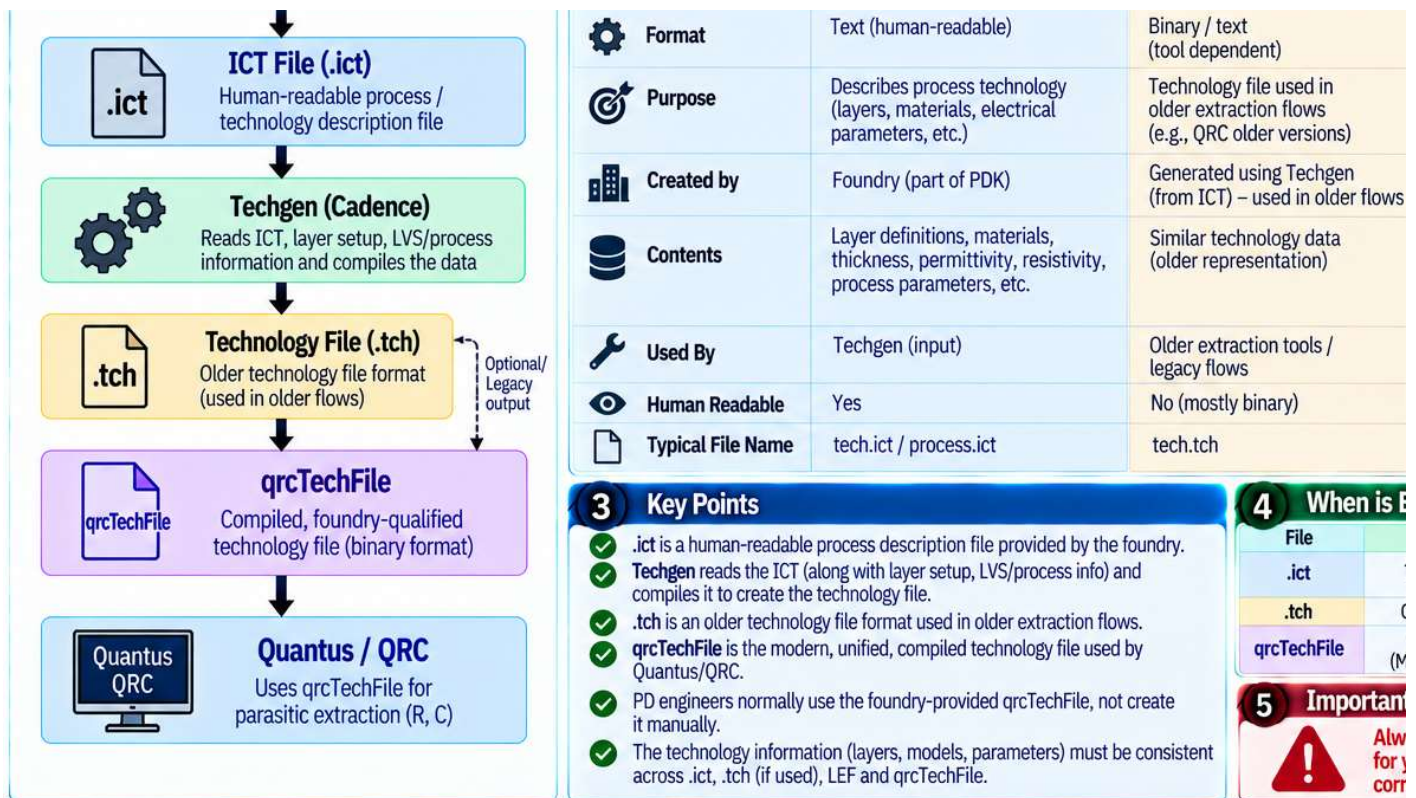
1 File Generation Flow



Foundry Process Data
(layers, materials, design rules, electrical parameters, etc.)

2 Comparison of .ict, .tch and qrcTechFile

Aspect	.ict	.tch
Full Form / Name	InterConnect Technology file	Technology file (older format)



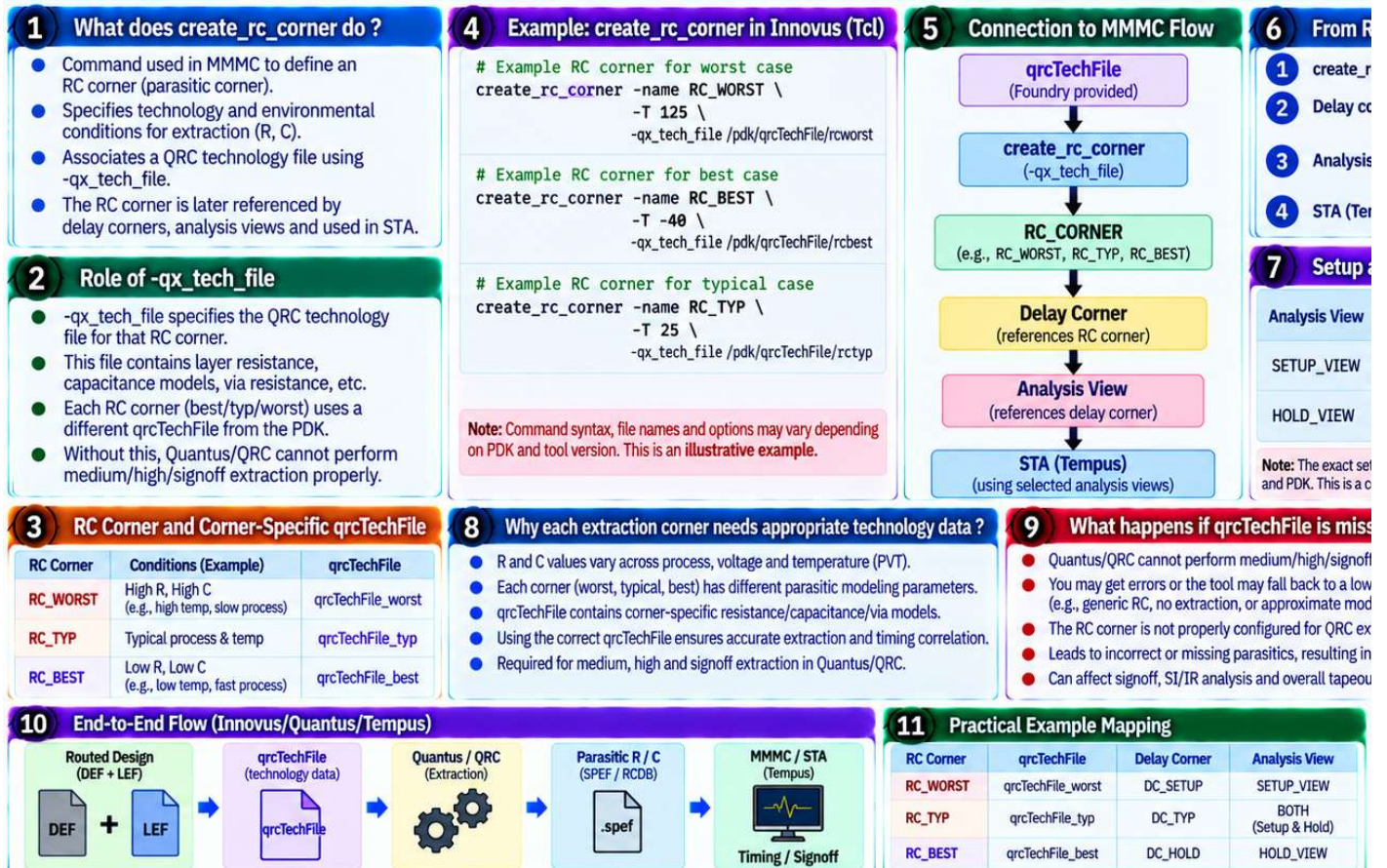
Takeaway:

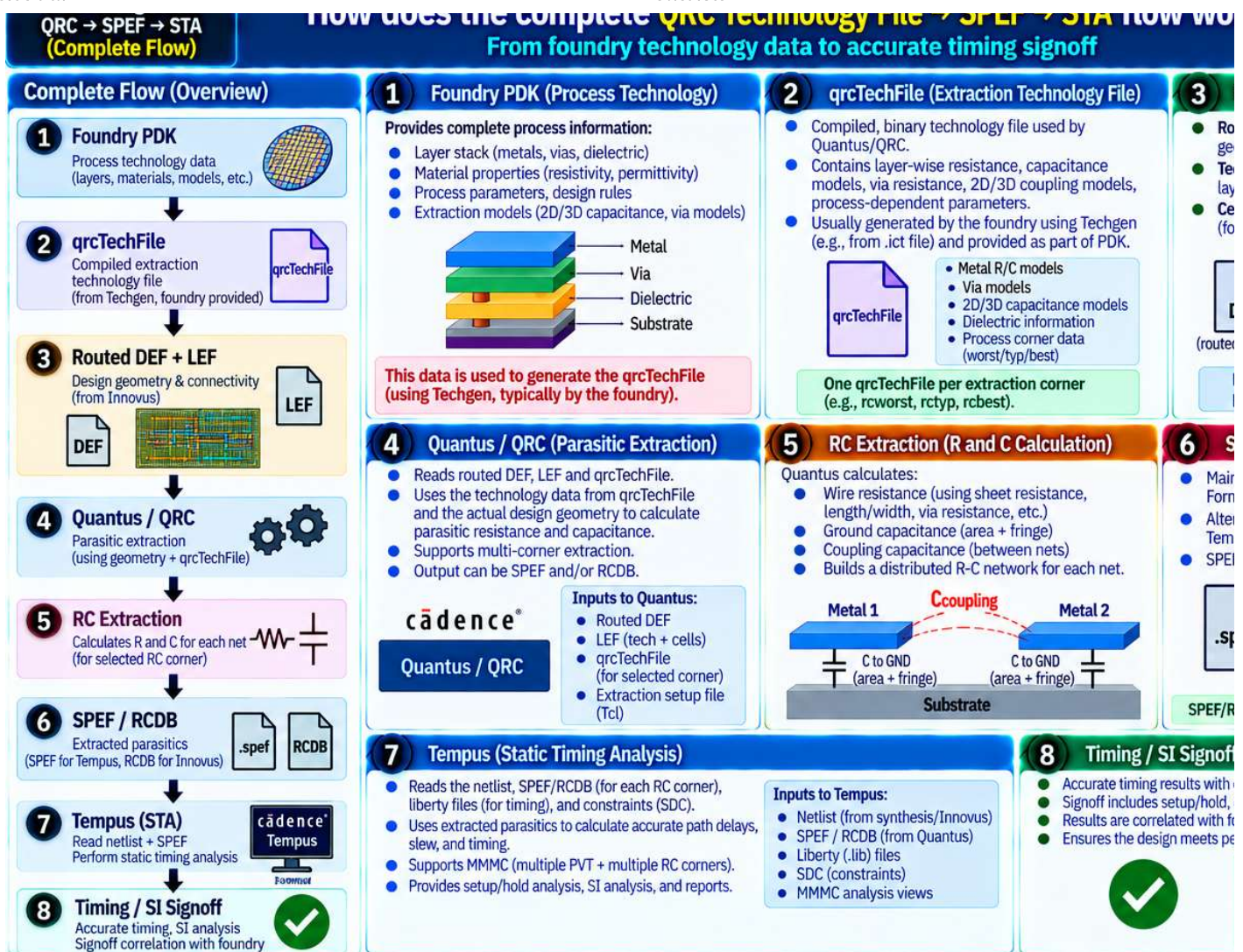
- Same technology information, different representation and usage.
- ICT → Techgen → (optional .tch) → qrcTechFile → Quantus Extraction.

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How is the QRC Technology File connected to MMC create_rc_corner

The qrcTechFile provides the extraction technology data, and MMC's create_rc_corner uses -qx_tech_file to associate that data with an RC corner, which is then used by delay corners, analysis views and STA.





Key Takeaways:

- qrcTechFile provides the technology (electrical) data.
- Routed DEF + LEF provide the physical geometry.
- Quantus uses both to extract R and C.

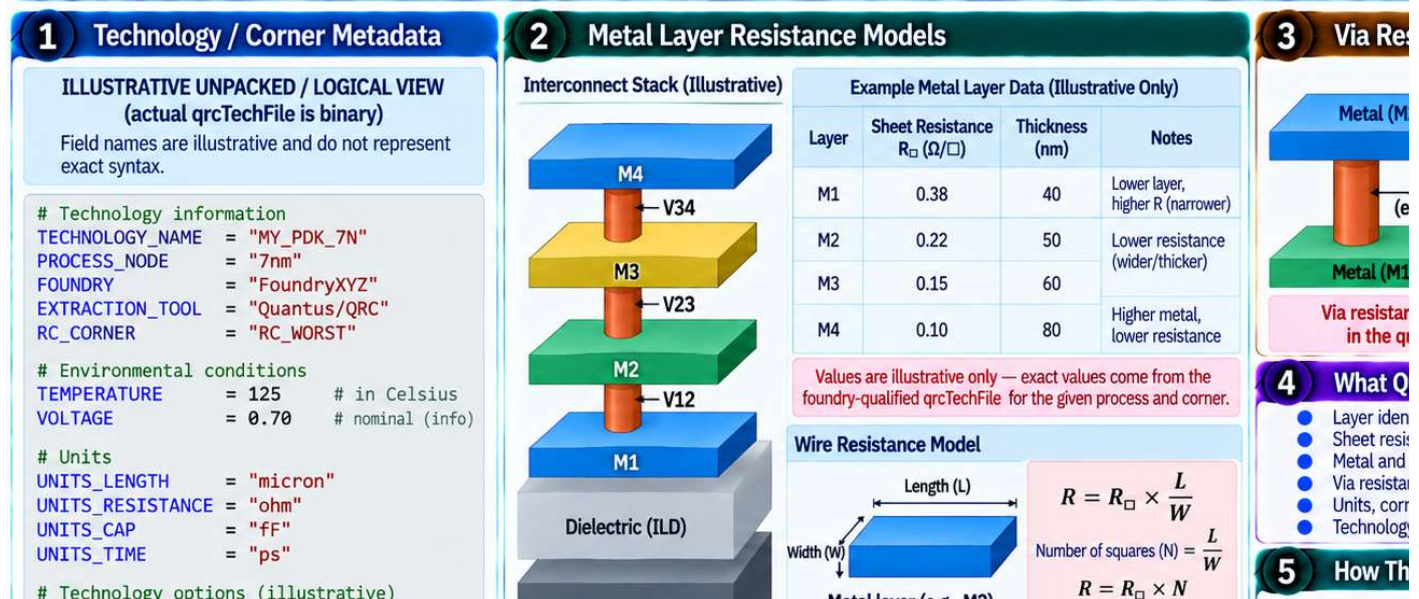
- SPEF/RCDB carries the extracted parasitics.
- Tempus uses the parasitics for accurate STA.
- Each RC corner needs the correct qrcTechFile.

- This flow ensures accurate timing and signoff correlation.

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Example of a .qrcTechFile — Part 1/3 Technology identity, units, layer definitions & resistance models

A unified qrcTechFile is a **binary technology file** provided/qualified by the foundry for use with Quantus. It contains process technology data including sheet resistivity information for conductor/diffusion layers, via resistances, and capacitance models.



```

DIELECTRIC_MODEL = "2D/3D"
INCLUDE_VIA_R    = "YES"
VERSION          = "1.0"
...

```

Substrate

R_Q = sheet resistance of the metal layer (from qrcTechFile)qrcTechFile
(technology data)

Part 1 Takeaway:

The **qrcTechFile** does not describe the routed design itself; it supplies the technology-dependent electrical models (layer/via resistance, etc.) that let Quantus turn DEF/LEF geometry into parasitic R/C.

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Example of a .qrcTechFile — Part 2/3

Dielectric definitions, layer stack, permittivity, and capacitance models (2D/3D)

2 .qrcTechFile Content (Part 2) — Dielectric & Capacitance Models

```

1 # =====
2 # Dielectric Materials and Layer Stack
3 # =====
4 MATERIAL OXIDE      TYPE = DIELECTRIC
5             PERMITTIVITY = 3.9      # SiO2
6 MATERIAL LOWK        TYPE = DIELECTRIC
7             PERMITTIVITY = 2.4      # low-k ILD
8 MATERIAL AIR          TYPE = DIELECTRIC
9             PERMITTIVITY = 1.0      # air (top)
10 MATERIAL SILICON     TYPE = SEMICONDUCTOR
11             PERMITTIVITY = 11.7    # substrate
12
13 # -----
14 # Layer Stack Definition
15 # -----
16 LAYER M1  MATERIAL = Cu    THICKNESS = 40n    ABOVE = ILD1
17 LAYER ILD1 MATERIAL = LOWK  THICKNESS = 100n   ABOVE = M1
18 LAYER M2  MATERIAL = Cu    THICKNESS = 50n    ABOVE = ILD2
19 LAYER ILD2 MATERIAL = LOWK  THICKNESS = 120n   ABOVE = M2
20 LAYER M3  MATERIAL = Cu    THICKNESS = 60n    ABOVE = ILD3
21 ...
22 # -----
23 # Capacitance Model Parameters (2D/3D)
24 # -----
25 CAP_MODEL      TYPE = "3D"      # options: 2D or 3D
26 FRINGE_MODEL   = "FIELD_SOLVER" # fringe capacitance model
27 COUPLING_MODEL = "FULL"         # FULL or TABLE
28 DIELECTRIC_THICKNESS_MODEL = "LAYER_BASED"
29 LATERAL_COUPLING = "ENABLED"    # lateral coupling between adjacent nets
30 VERTICAL_COUPLING = "ENABLED"   # coupling to adjacent layers
31 ...

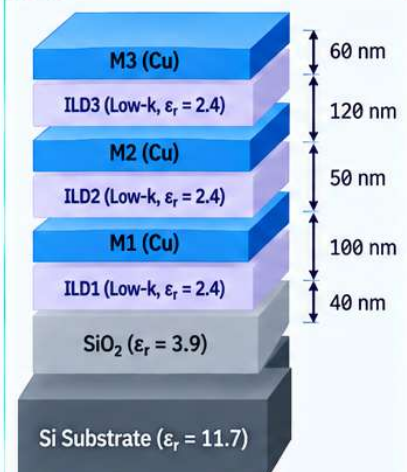
```

1 What This Section Contains ?

- Dielectric material definitions (e.g., SiO₂,
- Permittivity (ϵ_r) values for each material
- Layer stack information (metals and inter-
- Thickness of each layer
- Capacitance modeling options (2D/3D)
- Fringe and coupling capacitance models
- Parameters that control how Quantus cal

These parameters tell Quantus how the ph and how to model the capacitances between and between layers.

2 Layer Stack Illustration (Example)



Key Takeaway (Part 2):

- This section defines the dielectric environment and layer stack of the technology.
- Permittivity, thickness and capacitance models directly affect the extracted capacitance (C).
- These values are provided by the foundry and are process/corner specific.

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Example of a .qrcTechFile — Part 3/3

Via definitions, advanced parameters and closing section

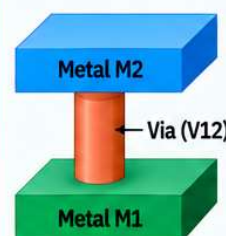
1 .qrcTechFile Content (Part 3) — Via Models, Advanced Options & End Section

```

1 # =====
2 # Via Definitions
3 # =====
4 VIA V12      = M1
5             LAYER1 = M2
6             RESISTANCE = 0.030      # via resistance in ohms
7             CAPACITANCE = 0.20f     # via to substrate capacitance
8             TYPE = "SINGLE"
9
10 VIA V23      =
11             LAYER1 = M2
12             LAYER2 = M3
13             RESISTANCE = 0.025
14             CAPACITANCE = 0.18f
15             TYPE = "SINGLE"
16 VIA V34      = ...

```

2 Via Structure and Parameters



Example Via Parameters

Parameter	Value (Example)
LAYER1	M1
LAYER2	M2
RESISTANCE	0.030 Ω
CAPACITANCE	0.20 fF
TYPE	SINGLE / ARRAY

Via resistance and capacitance are technology dependent and provided in the **qrcTechFile** for each via type.

```

17 LAYER1      = M3
18 LAYER2      = M4
19 RESISTANCE   = 0.020
20 CAPACITANCE = 0.15f
21 TYPE        = "SINGLE"
22 ...
23 # =====
24 # Advanced Extraction / Modeling Options
25 # =====
26 EXTRACTION_DIMENSION = "3D"          # 2D or 3D
27 FRINGE_CAP_MODEL      = "TABLE"       # TABLE or FIELD_SOLVER
28 COUPLING_DISTANCE     = 5n           # minimum coupling distance
29 DIELECTRIC_THICKNESS_MODEL = "LAYER_BASED"
30 SUBSTRATE_MODEL       = "RC"         # substrate modeling
31 TEMPERATURE_COEFFICIENT = "ENABLED"  # R/C variation with temp
32 PROCESS_VARIATION     = "ENABLED"    # for multi-corner extraction
33 QRC_ACCURACY          = "HIGH"       # LOW / MEDIUM / HIGH
34 ...
35 # =====
36 # End of File
37 # =====
38 END

```

4 Advanced Options Explained

5

Parameter	Meaning
EXTRACTION_DIMENSION	2D (faster) or 3D (more accurate)
FRINGE_CAP_MODEL	How fringe capacitance is calculated (TABLE or field solver based)
COUPLING_DISTANCE	Minimum distance to consider coupling capacitance
SUBSTRATE_MODEL	Substrate resistance/capacitance model (e.g., RC, analytical)
TEMPERATURE_COEFFICIENT	Enables R/C variation with temperature
PROCESS_VARIATION	Enables multi-corner extraction (worst/typ/best)
QRC_ACCURACY	Controls extraction accuracy vs. runtime

These options control how Quantus calculates parasitics using the technology data. The right settings are provided by the foundry.



Key Takeaways (Part 3):

- Defines via resistance and capacitance models for each via type.
- Provides advanced extraction options (2D/3D, fringe, coupling, substrate, variation).
- Completes the technology file with end of file section.

